data processing). Similarly, blockchain programming is yet another paradigm shift.

Four fundamental concepts play a significant role in making blockchain programming different. You need to understand these concepts before you start programming in chapter 2, just as you need to learn about class and object concepts before undertaking OOP (object-oriented programming). Given this context, the four key roles fulfilled by a blockchain are

- *Decentralized infrastructure* --Special computing hardware and software stacks support the blockchain protocol, smart contracts, and applications (Dapps). The main components of this infrastructure are the computing nodes and networks connecting the nodes (section 1.3.1).
- *Distributed ledger technology* --On top of the infrastructure is the ledger. Transactions and data are recorded simultaneously in all stakeholders' ledgers. The ledger is *distributed* because all the stakeholders record the same facts. It is *immutable* because each block is linked to the signature of the previous block, making it tamperproof (section 1.3.2).
- *Disintermediation protocol* --Participants in a decentralized system follow the same blockchain protocol to connect and to communicate and transact with one another. The protocol is a set of rules for everyone to follow. Ethereum and Hyperledger, for example, are two different blockchain protocols (section 1.3.3).